

Preprocessing and EDA

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Overview

This report outlines the exploratory data analysis (EDA) and preprocessing steps performed on the "Messy Mashup" dataset. The analysis focuses on validating dataset integrity, identifying silent anomalies in the audio tracks using `librosa.effects.split`, and establishing a baseline for mixing multi-stem audio files.

Question-Wise Analysis

Q1. What is the value of total number of corrupted sounds (less than 4kb) + (total number of sounds < 5.0491MB)?

Approach: Iterated through the 10 genre directories and their subfolders using `os.path.getsize()` to compute file sizes in bytes, tallying corrupted files (< 4KB) and files falling below the specified MB threshold.

Answer: 1256

Q2. What is the absolute difference between total number of sounds > 5.0493MB and total number of sounds < 5.0491MB?

Approach: Maintained parallel counters during the directory traversal to track the exact file size distribution thresholds, then computed the absolute difference.

Answer: 1072

Q3. What is the absolute difference between the number of training reggae drum samples and the number of validation country vocal samples?

Approach: Filtered out corrupted tracks and split the valid dataset using a fixed random seed (83/17 split for train/validation). Calculated the absolute difference in dictionary list lengths between the specified subsets.

Answer: 66

Q4. Total number of sound files having silence greater than equal to 5 secs.

Approach: Loaded audio tracks using `librosa.load` at 22050 Hz. Extracted non-silent intervals using `librosa.effects.split(y, top_db=20)` and mathematically calculated the maximum duration (gaps) between these intervals.

Answer: 676

Q5. Total number of sound tracks in Vocals where silence ≥ 5 secs.

Approach: Filtered the generated silence dataframe (from Q4) specifically for rows where the 'Stem' column equaled 'vocals'.

Answer: 304

Q6. What's the average Silence Length in Vocals (in secs)?

Approach: Computed the statistical mean of the `Max_Silence_Sec` column for the filtered vocals dataframe.

Answer: 12.79

Q7. Total number of drums sound tracks in jazz where silence ≥ 5 secs.

Approach: Filtered the overall silence dataframe to isolate records where the genre was 'jazz' and the stem was 'drums'.

Answer: 21

Q8. Total number of drums sound tracks in jazz where silence ≥ 5 secs and Silence_Location is only middle.

Approach: Further filtered the jazz/drums subset to count instances where the gap calculation logic flagged the silence exclusively as 'middle'.

Answer: 15

Q9. Total number of drums sound tracks in jazz where silence ≥ 5 secs and Max_Silence_Sec ≥ 10 .

Approach: Filtered the jazz/drums subset to check for tracks containing extreme silent gaps exceeding 10 seconds.

Answer: 7

Audio Mixing and Normalization

For the following questions, the first song from the 'rock' genre was selected. Exactly 5.0 seconds of all four stems (`drums`, `vocals`, `bass`, `other`) were loaded, stacked into a NumPy array, and combined via element-wise addition (`np.sum(stems_stack, axis=0)`) to create a raw mix.

Q10. What is the length of the mix sample?

Approach: Checked the total number of items in the resulting 1D NumPy array using `len()`, derived from 5.0 seconds at a 22050 Hz sample rate.

Answer: 110250

Q11. What is the value of RMS Amplitude of mix sample?

Approach: Calculated the Root Mean Square mathematically to measure

the average power of the mixed signal using the standard formula: $RMS =$

$$\sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}.$$

Answer: 0.11

Q12. What is the value of max value of peak normalized sample?

Approach: Performed peak normalization by dividing the raw mix array by its absolute maximum value (`np.max(np.abs(mix_raw))`), then extracted the maximum positive peak of this normalized array.

Answer: 0.89